

Exogenic τ -Cascade: A Falsifiable ENT-Aligned Hypothesis with Reproducible Evidence

Abstract

We formalize the *Exogenic τ -Cascade* (XTC) as a falsifiable hypothesis within Emergent Necessity Theory (ENT). ENT models coherence via the ratio $\tau(t) = E_{\text{syn}}(t)/E_{\text{drift}}(t)$. An external perturbation $S_{\text{ext}}(t)$, coupled by susceptibility χ and coupling γ , injects drift energy and can depress τ below a coherence threshold τ_c , with propagation across coupled subsystems. We state falsifiability criteria and containment levers, relate XTC to established complex-systems notions (cascading failure, tipping points, exogenous shock propagation), and provide three minimal, fully reproducible simulations (network synchrony, queue instability, and symbolic recursion). The paper makes no claims beyond those conditions and invites refutation.

1 Introduction

External shocks often precipitate cascades in power grids, markets, traffic, and symbolic processes. These are typically described as *cascading failures*, *tipping points*, or *shock propagation*. ENT contributes a structural lens by tracking the coherence ratio

$$\tau(t) = \frac{E_{\text{syn}}(t)}{E_{\text{drift}}(t)}, \quad (1)$$

where E_{syn} sustains recursion and E_{drift} erodes it. The *Exogenic τ -Cascade* (XTC) hypothesis states that $S_{\text{ext}}(t)$ can couple into a system through (γ, χ) to depress $\tau(t)$ past a threshold τ_c , initiating a cascade across connected subsystems.

2 Formalization (ENT)

Let $E_{\text{drift}}^0(t)$ denote drift absent shocks. Under an external driver:

$$E_{\text{drift}}(t) = E_{\text{drift}}^0(t) + \gamma(\chi * S_{\text{ext}})(t), \quad (2)$$

with convolution $*$ for temporal susceptibility χ . A cascade is *structurally* defined as

$$\exists t \in [t_0, t_0 + \Delta t] : \tau(t) \leq \tau_c \quad \text{and} \quad \frac{d\tau}{dt} < 0 \text{ pre-crossing}, \quad (3)$$

followed by propagation to coupled subsystems. We do not claim inevitability; only that these measurable conditions can occur.

3 Relation to Established Concepts (non-ENT readers)

XTC is not a new label for cascades; it is an ENT framing that unifies shock, susceptibility, and coherence into one falsifiable ratio.

- **Cascading failure** (networks): local disruption triggers global collapse.
- **Tipping points**: threshold-crossing dynamics.
- **Shock propagation**: exogenous stress transmitted along couplings.

XTC's contribution is the explicit ratio τ linking shock energy, susceptibility, and drift to thresholded behavior testable across domains.

4 Falsifiability & Containment

4.1 Falsifiability criteria

Reject XTC if any hold:

1. No measurable $\Delta\tau < 0$ under S_{ext} .
2. No dose-response in (γ, χ) on onset timing or crossing probability.
3. Apparent cascades occur independently of $S_{\text{ext}}(t)$.

4.2 Containment levers (ENT)

- **Shielding/decoupling**: $\downarrow \gamma$.
- **Damping/filters**: $\downarrow \chi$.
- **Redundancy/feedback**: $\uparrow E_{\text{syn}}$.

5 Operational mappings for simulations

The ENT definition of τ is general. For reproducibility, each testbed uses a *clear operational mapping* with $\tau_c = 1.0$:

- (a) **Network synchrony (Kuramoto)**: order parameter $R(t) \in [0, 1]$; $E_{\text{syn}} = R$, $E_{\text{drift}} = 1 - R$, so $\tau = R/(1 - R)$. Thus $R = 0.5 \Leftrightarrow \tau = 1$.
- (b) **Queue instability (M/M/1)**: service μ , effective arrival $\lambda_{\text{eff}}(t)$; $E_{\text{syn}} = \mu$, $E_{\text{drift}} = \lambda_{\text{eff}}$, so $\tau = \mu/\lambda_{\text{eff}}$. Thus $\lambda_{\text{eff}} = \mu \Leftrightarrow \tau = 1$.
- (c) **Symbolic recursion surrogate** ($s(t) \in [0, 1]$): $E_{\text{syn}} = s$, $E_{\text{drift}} = 1 - s$, so $\tau = s/(1 - s)$. A nonlinear susceptibility is used to reflect increased brittleness as coherence falls.

6 Simulation evidence (reproducible)

All code and CSV outputs are in the ENT repository under `simulations/exogenic_tau_cascade/` (scripts, seeds, and parameters in the README). The external shock window and absolute times are specified per test.

6.1 Kuramoto network (shock window $t \in [10, 30]$ s)

$N=50$, $K=1.2$, incoherent forcing applied to 22 nodes with random phases; pre-shock state is synchronized ($\tau > 1$). Results (`kuramoto_summary.csv`):

- $\gamma=0.10, 0.20, 0.30$: no crossing ($\tau > 1$ throughout shock).
- $\gamma=0.35$: crossing at $t \approx 12.7$ s (shock starts at 10 s).
- $\gamma=0.40$: crossing at $t \approx 11.8$ s.

These show dose-response in γ and onset within a short window after shock onset.

6.2 M/M/1 queue with burst (shock window $t \in [60, 120]$ s)

Baseline $\lambda_0=0.75$, $\mu=1.0$, burst $\lambda_{\text{eff}}=\lambda_0 + \gamma A$ with $A=0.7$. Results (`queue_summary.csv`):

- $\gamma=0.20, 0.30, 0.35$: no crossing; $\tau=\mu/\lambda_{\text{eff}} > 1$.
- $\gamma=0.40, 0.50$: $\lambda_{\text{eff}} \geq \mu$, immediate crossing at $t=60$ s.

This explicitly demonstrates the threshold condition $\lambda_{\text{eff}} = \mu \Leftrightarrow \tau = 1$ and a clean dose-response in γ .

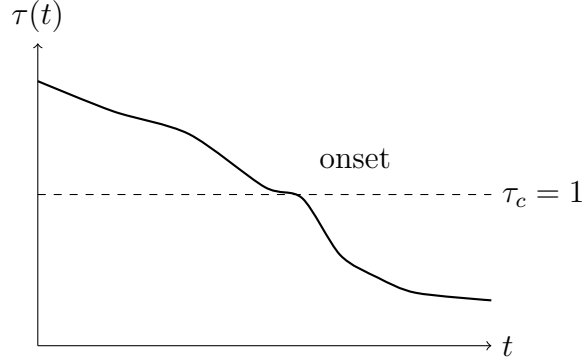
6.3 Symbolic recursion surrogate (shock window $t \in [60, 200]$)

State $s \in [0, 1]$ with nonlinear susceptibility (exponent $p=1.5$) and stronger feedback once $s < 0.6$; noise is small and seed is fixed. Results (`symbolic_summary.csv` and `symbolic_containment_gamma_0.`

- $\gamma=0.10$: $\approx 33\%$ reduction in τ at end of shock, *no crossing*.
- $\gamma=0.30$: τ crosses below 1 at $t \approx 139$ s.
- **Containment:** at $\gamma=0.30$, reducing susceptibility to $\chi=0.5$ prevents crossing (time series shows $\tau > 1$ for the entire window).

This test isolates reversibility via χ (damping), consistent with ENT levers.

7 Illustrative trajectory (generic)



8 Discussion (scope and justification)

The term *Exogenic τ -Cascade* is not a standard label in the literature; it is an ENT framing that makes three things explicit: (i) a measurable ratio τ tied to coherence vs. drift; (ii) a shock pathway $S_{\text{ext}} \rightarrow (\gamma, \chi)$; and (iii) thresholded, falsifiable onset with containment levers. The presented simulations are minimal and domain-agnostic: they demonstrate *correlations and threshold behavior*, not new laws. The hypothesis is thus both constrained (no overreach) and empirically refutable.

Data and Code Availability

All scripts and deterministic CSV outputs are in the ENT repository under:

```
simulationa/exogenic_tau_cascade/
```

with a concise README describing mappings, parameters, and measurement windows. Files include per- γ time series and summaries for each test.

References

- [1] Dobson, I. et al. (2007). Complex systems analysis of series of blackouts: cascading failure, critical points, and self-organization. *Chaos*, 17(2).
- [2] Scheffer, M. et al. (2009). Early-warning signals for critical transitions. *Nature*, 461(7260), 53–59.